Table 1: Does the density objective improve HILP or RLDP? OGBench manipulation (proprioceptive input), success rate (%), reward-inference eval, 1M steps. "lin" = density retrofit with the linear actor reward $\phi(u)^\top z$ (campaign code); "lin, fixed" = same, retrained on the repaired retrofit code; "exact" = density retrofit whose actor maximises the exact goal-kernel reward on the candidate states (full-state candidate input). Plain HILP / RLDP are our reimplementations under the identical protocol; Density-FB is the deployed z model.

	HILP backbone				RLDP backbone				Density-FB
	plain	D lin	D lin, fixed	D exact	plain	D lin	D lin, fixed	D exact	z^{R}
cube-single	71.1 ±5.7	71.1 ±4.8	67.5 ±10.3	74.0 ±5.7[f2]	25.0	—	23.8 ±6.7	30.0 ±0.0[f2]	81.3 ±3.8
cube-double	11.1 ±3.3	2.4 ±1.6	—	2.0 ±0.0[f2]	2.2 ±1.1	2.0 ±0.5	—	1.0 ±1.4[f2]	30.4 ±2.6
scene	39.3 ±5.7	42.0 ±0.9	44.0 ±2.4[‡]	16.7 ±1.2[f]	4.7 ±1.1	—	6.8 ±2.1[‡]	2.7 ±3.1[f]	52.7 ±3.3
puzzle-3x3	1.3 ±0.5	2.2 ±1.3	—	0.7 ±1.2[f]	2.4 ±1.1	—	—	0.0 ±0.0[f]	4.7 ±0.9

All cells: state input, flow actor, 1M steps, reward-inference eval, last-3-checkpoint estimator, mean ± std over 3 seeds, except: [‡] seed 2 at 600k when tabulated; [f] final-step estimator (luis gk arm, receipts carry the final eval only); [f2] final-step, 2 seeds. "plain" HILP = hilp_tdjc, RLDP = rldp; "D lin" = hilp_density_tdjc / rldp_density (campaign code, commit 6cc2952); "D lin, fixed" = same critics retrained 2026-09-11 on the repaired retrofit code (RLDP reward-coefficient least-squares solve, HILP temperature / min-head option plumbing, actor divide-by-zero guard, object-goal predicate); "D exact" = hilp_density_gk / rldp_density_gk (goal_kernel=true, full-state candidate tower). Reading: with the host method's own rule for choosing the task vector, the density objective inside HILP or RLDP is level with the plain backbone on every cell, and adding the exact-reward actor hurts on scene (HILP 42→17) and does nothing elsewhere. The objective does transfer once the task vector comes from a hindsight goal and the reward regression instead: see Table **??**, where RLDP goes 28 → 53 and HILP 71 → 77 on cube-single at full state. Goal-conditioned route for the retrofits (valid only on the repaired code): D-HILP lin, fixed cube-single 36.4±3.2, scene 51.8±3.5; D-RLDP lin, fixed cube-single 21.4±3.2, scene 7.2±0.2.